

EE115A Analog Circuits

Homework 1

Due on 23: 59, October 1, 2026.

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Please use English only to complete the assignment.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Exercise 1: Circuit Basics (10pt)

For the circuit in Figure 1, find the Thévenin equivalent circuit between terminals (a) 1 and 2, (b) 1 and 3.

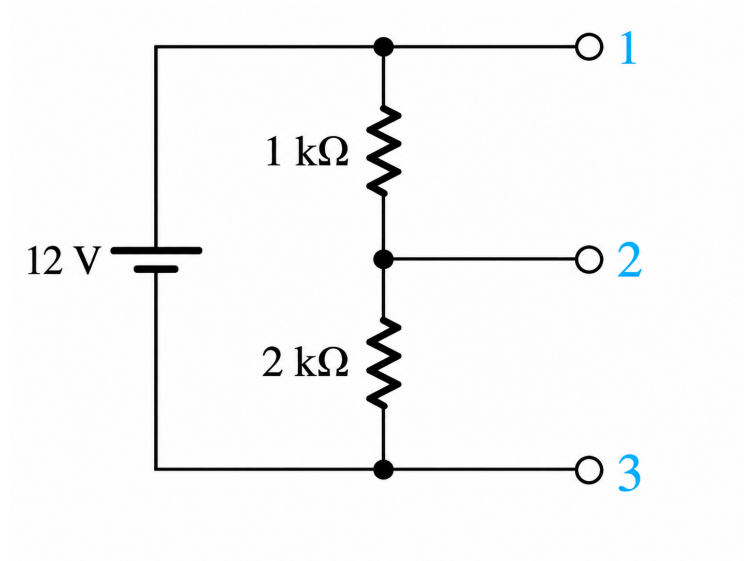


Figure 1: Circuit for exercise 1.

Exercise 2: Frequency Response (20pt)

Figure 2 shows a signal source connected to the input of an amplifier. Here R_s is the source resistance, and R_i and C_i are the input resistance and input capacitance, respectively, of the amplifier.

(a) Derive an expression for

$$\frac{V_i(s)}{V_s(s)},$$

and show that it is of the low-pass STC type.

(b) Find the 3-dB frequency for the case

$$R_s = 10 \text{ k}\Omega, \quad R_i = 10 \text{ k}\Omega, \quad C_i = 10 \text{ pF}.$$

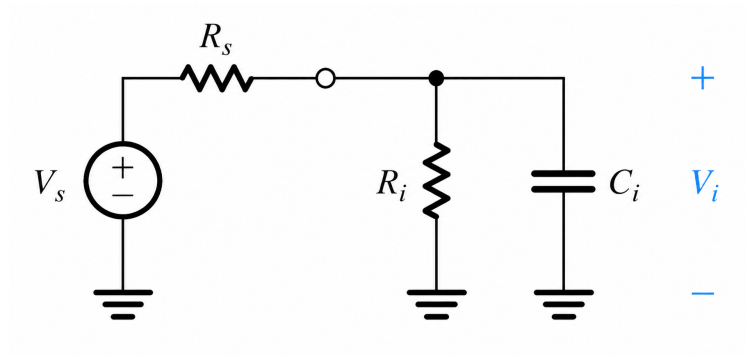


Figure 2: Circuit for exercise 2.

Exercise 3: Circuit Model (20pt)

Figure 3 shows a transconductance amplifier whose output is fed back to its input.

- (a) Find the input resistance R_{in} of the resulting one-port network. (Hint: Apply a test voltage v_x between the two input terminals, and find the current i_x drawn from the source.)
- (b) Show that when driven by a voltage source v_s , having source resistance

$$R_s = R_{\text{in}},$$

the voltage gain

$$\frac{v_o}{v_s} = 0.5 \frac{v_o}{v_i}.$$

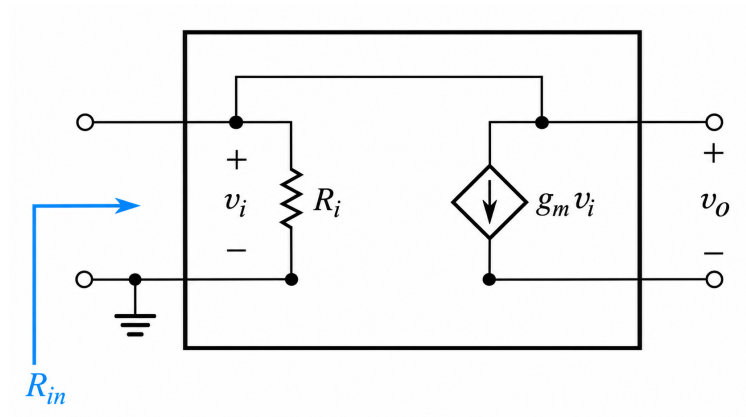


Figure 3: Circuit for exercise 3.

Exercise 4: Amplifiers 1 (30pt)

Assuming ideal op amps, find the voltage gain v_o/v_i and input resistance R_{in} of each of the circuits in Figure 4.

Note: This question requires simulation. Please note that you need to use **Multisim** for simulation and provide the simulation circuit and simulation results for **circuit (a)** and **circuit (c)** in your answer.

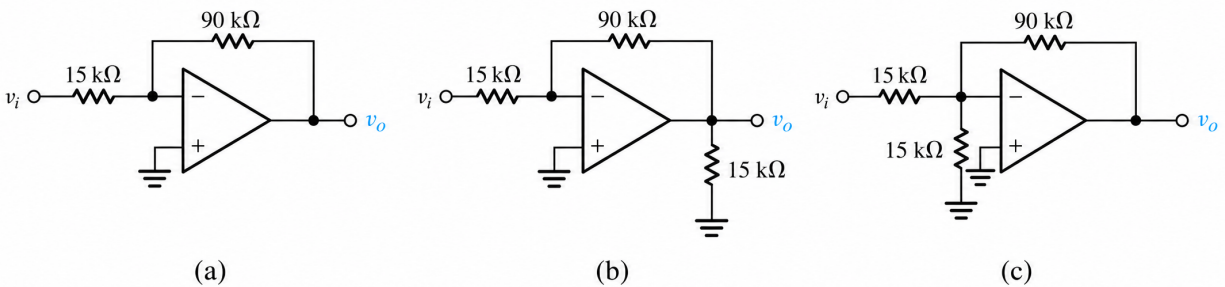


Figure 4: Circuit for exercise 4.

Exercise 5: Amplifiers 2 (20pt)

For the circuit shown in Figure 5, use the superposition principle to determine the output voltage v_o in terms of the input voltages v_1 and v_2 . Assume an ideal op amp. For

$$v_1 = 5 \sin(2\pi \times 100t) - 0.2 \sin(2\pi \times 2000t) \text{ V},$$

$$v_2 = 5 \sin(2\pi \times 100t) + 0.2 \sin(2\pi \times 2000t) \text{ V}.$$

Find v_o .

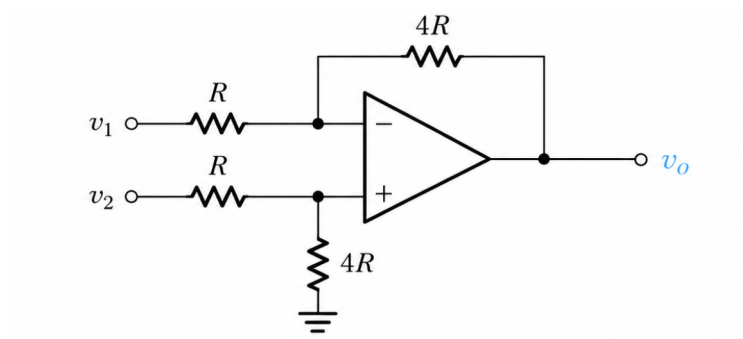


Figure 5: Circuit for exercise 5.